

Methods 3: Multilevel Statistical Modeling and Machine Learning

Week 38: *Multilevel linear regression*
September 15, 2026

The course plan

Week 37:

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model

Week 38:

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects?

Week 39:

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous?

Week 40:

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review

Week 41:

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models

Week 42:

no teaching (go fetch potatoes instead)

Week 43:

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running

Week 44:

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses

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Lesson 7: Model evaluation and hyperparameter tuning

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Week 46:

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data?

Week 47:

Lesson 9: Neural networks

Class 9: Code review

Week 48:

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource)

Week 49:

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning

Week 50:

Student presentations:

Class 12: Work on portfolios: Ask anything!

Learning goals and outline

Multilevel linear regression

- 1) Understanding the motivations for doing multilevel modelling
 - Using all the data
 - Respecting the data distribution
- 2) Understanding the basics of multilevel modelling
 - Different kinds of design matrices, X and Z
 - Variance-covariance matrices
 - Pooling: complete pooling, no pooling and partial pooling

Overall motivation

We want to use all the information in the data while respecting the data distributions the data is generated from

The four classical levels of variables

- Nominal
 - examples: true/false, correct/incorrect, female/male, dog/cat, apples/pear, also called *categorical*
 - they are **names** of categories, but it does not make sense to order them
- Ordinal
 - examples: senior/junior, adult/child 1/2/3
 - they are also **names** of categories, but there is an explicit or implicit **ordering**, i.e. one is greater than another
- Interval
 - examples: the year 1984 AD; the temperature 100 °C
 - they can be **continuous**, and there is **ordering**, e.g. 100 °C > 90 °C. And intervals can be compared, e.g. the interval from 80 °C to 100 °C is as long as the one from 40 °C to 60 °C
 - crucially, there is no real 0; the year before 1 AD is not characterised by absence of time; and 0 °C is not characterised by the absence of temperature. This means that we cannot say that, say, 40 °C is twice as high a temperature as 20 °C
- Ratio
 - examples: the temperature 273 K, the reaction time of a subject
 - they can be **continuous**, there is **ordering** and there is a **real 0**.
 - Thus we can say that 200 K is twice the temperature of 100 K, as 0 K *is* the absence of temperature; and we can say that subject 2, 400 ms, is twice as fast as subject 1, 200 ms, because 0 ms *is* the time when the event happened

Sub-optimal practices

TREATING VARIABLES AS ANOTHER LEVEL THAN THEY ARE



sleepstudy {lme4}

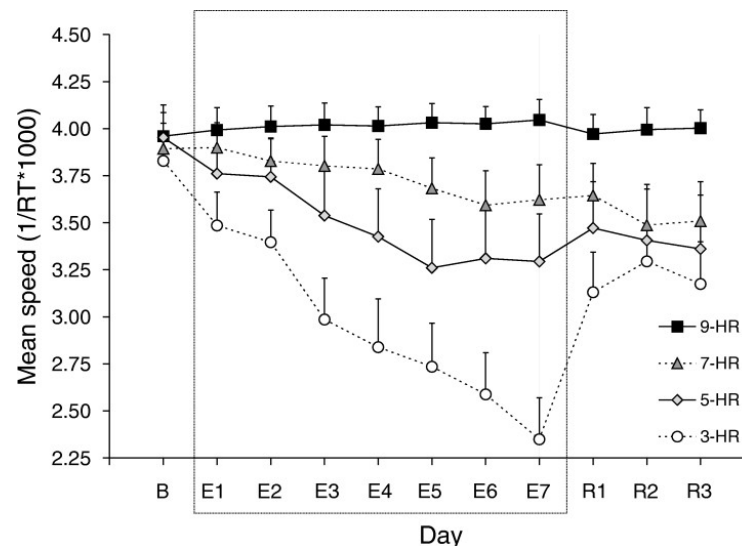
R Documentation

Reaction times in a sleep deprivation study

Description

The average reaction time per day (in milliseconds) for subjects in a sleep deprivation study.

Days 0-1 were adaptation and training (T1/T2), day 2 was baseline (B); sleep deprivation started after day 2.



Q: judging from the figure: what kind of predictor is *Day* modelled as?
And should it have been modelled otherwise?

Sub-optimal practices

AGGREGATION – part 1



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Journal of Experimental Psychology: General

<https://doi.org/10.1037/xge0001091>

Effects of Statistical Learning in Passive and Active Contexts on Reproduction and Recognition of Auditory Sequences

Saloni Krishnan¹, Daniel Carey², Frederic Dick^{3, 4}, and Marcus T. Pearce^{5, 6}

¹ Department of Psychology, Royal Holloway, University of London

² Novartis Pharmaceuticals, Dublin, Ireland

³ Department of Psychological Sciences, Birkbeck, University of London

⁴ Department of Experimental Psychology, University College London

⁵ School of Electronic Engineering and Computer Science, Queen Mary University of London

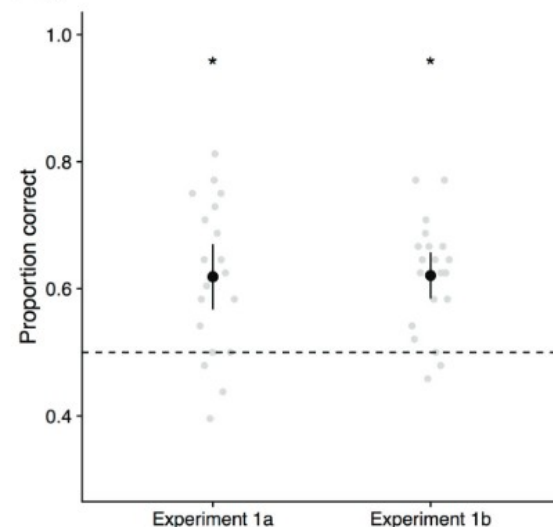
⁶ Department of Clinical Medicine, Aarhus University

Q's:

- What do each of the grey dots represent?
- What kind of response data is present at the single-trial level?
 - And how is the response data categorised here?
- Within a single-subject level framework, what would be the appropriate model to fit?
- Can confidence intervals extend beyond proportion correct 1 or 0 with the approach used here?

Figure 3

Mean Proportions of Correct Responses for the Recognition Task in Experiments 1a (Familiarization Only) and 1b (Reproduction Only)



Note. Chance performance is at 0.5, shown by the horizontal line, and asterisks indicate performance differing significantly from chance. Error bars represent 95% confidence intervals around the mean.

Sub-optimal practices

AGGREGATION – part 2



Cognitive Brain Research
Volume 7, Issue 4, March 1999, Pages 493-501



Research report

'Paradoxical' alpha synchronization in a memory task

W. Klimesch, M. Doppelmayr, J. Schwaiger, P. Auinger, Th. Winkler

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[https://doi.org/10.1016/S0926-6410\(98\)00056-1](https://doi.org/10.1016/S0926-6410(98)00056-1)

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“The percentage of epochs that were excluded from data analysis (due to artifacts) for hits and correct rejections ranged from 28% to 42% in the four experimental conditions.” section 2.4.2

“Three factorial ANOVA's were calculated for hits and correct rejections [...]. For these [electrode] sites, data were averaged separately for the left and right hemisphere but only for those 1 s intervals of an epoch which represent the first 1000 ms after presentation onset of the memory set and frame.” section 2.4.5

Q: What may be a problem when data is aggregated like that?

General Linear Model

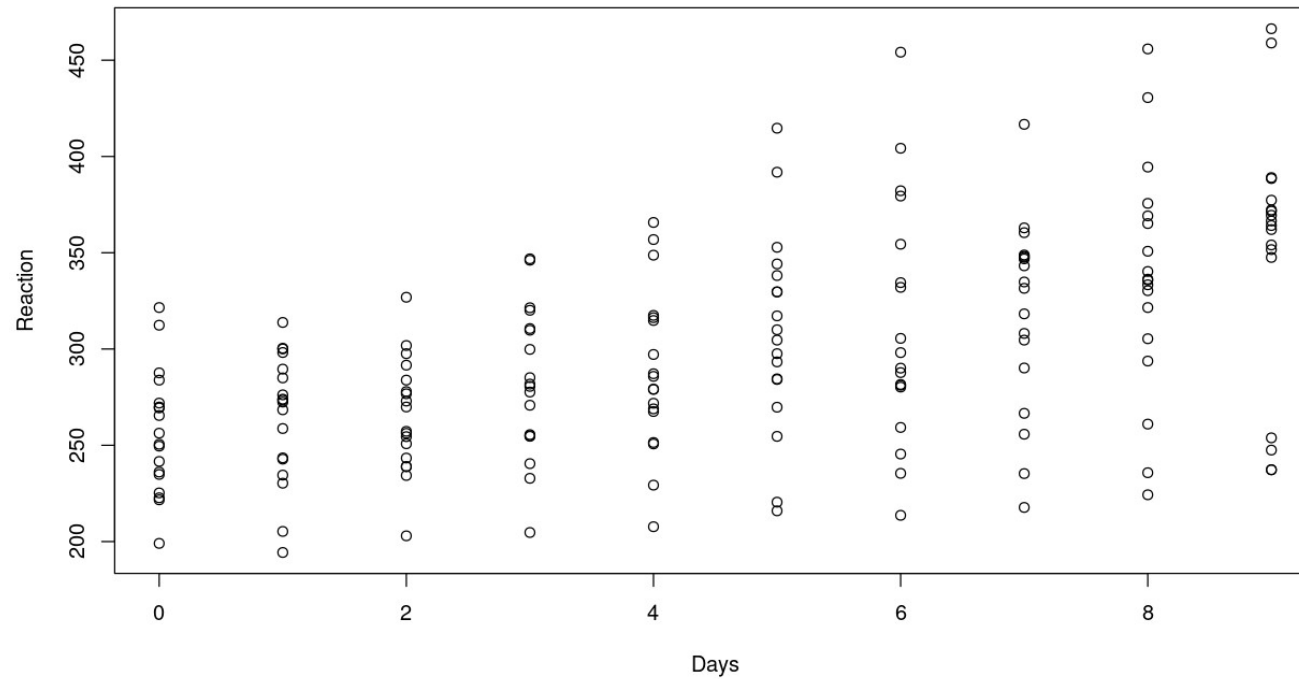
When all you have is a ~~hammer~~, everything looks
like a ~~nail~~

**Normally distributed variable
(an interval or ratio variable)**

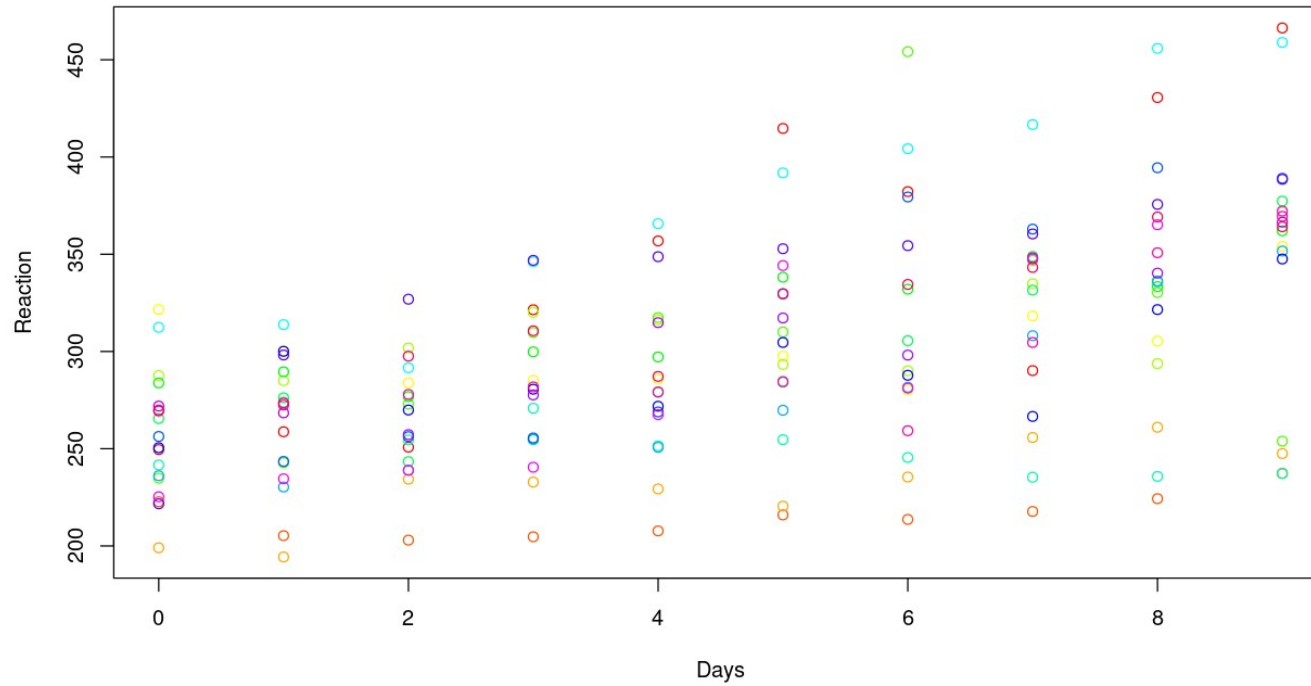
W38_live_coding

Understanding the basics of multilevel modelling

Sleep Study



Sleep Study



Design matrices

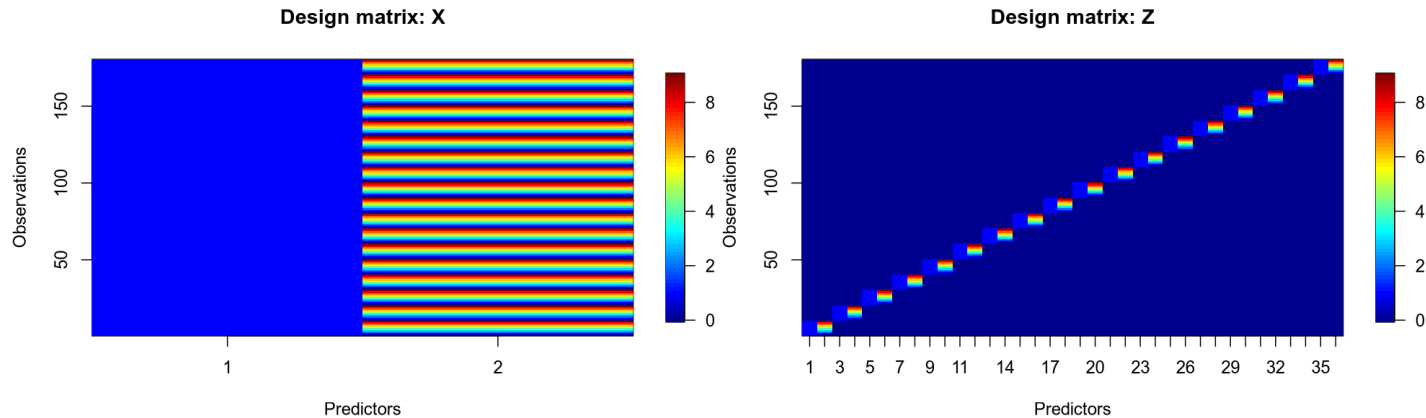
Level 1

Level 2

$$Y = X\beta + Zu + \epsilon$$

Fixed
effects

Random
effects



Linear mixed model fit by REML ['lmerMod']
 Formula: Reaction ~ Days + (Days | Subject)

Data: sleepstudy

REML criterion at convergence: 1743.628

Random effects:

Groups	Name	Std.Dev.	Corr
Subject	(Intercept)	24.741	
	Days	5.922	0.07

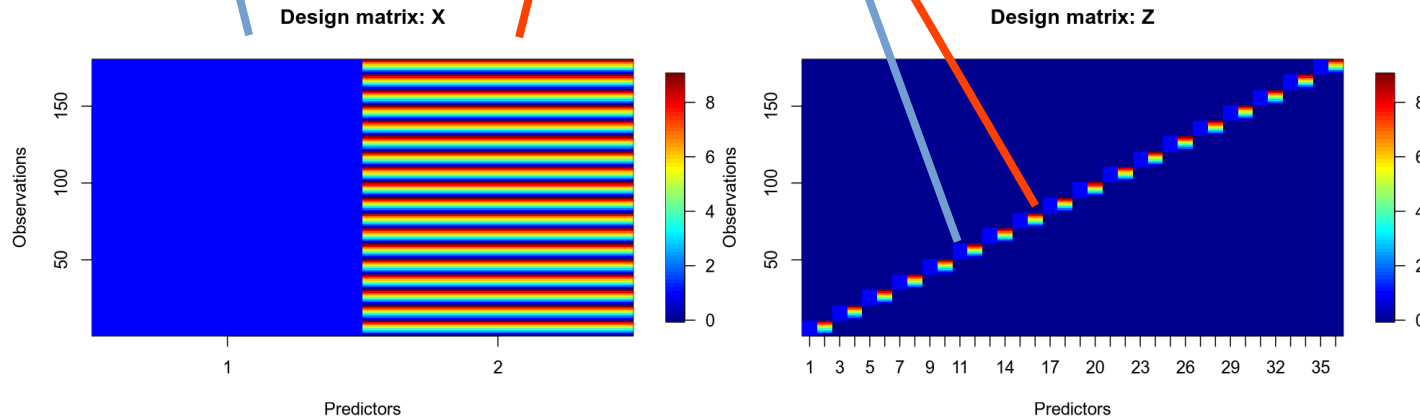
Residual 25.592

Number of obs: 180, groups: Subject, 18

Fixed Effects:

(Intercept)	Days
251.41	10.47

$$\hat{\sigma}; \epsilon \sim N(0, \sigma^2)$$



Variance-covariance matrices

Variance-covariance matrix

$$u \sim N(0, \Sigma) \quad (3)$$

$$\Sigma_{\theta} = \sigma^2 \Lambda_{\theta} \Lambda_{\theta}^T \quad (4)$$

$$\mathbf{G} = \Lambda_{\theta} \Lambda_{\theta}^T$$

Bates D, Mächler M, Bolker B, Walker S (2015) Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software 67:1–48. <https://doi.org/10.18637/jss.v067.i01>

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Residual 25.592

Number of obs: 180, groups: Subject, 18

Fixed Effects:

(Intercept)	Days
251.41	10.47

$$u \sim N(0, \Sigma)$$

$$\hat{\Sigma}_{(1:2, 1:2)} = \begin{pmatrix} 24.741^2 & 9.61 \\ 9.61 & 5.922^2 \end{pmatrix}$$

$$\mathbf{G}_{(1:2, 1:2)} = \begin{pmatrix} 0.93 & 0.015 \\ 0.015 & 0.054 \end{pmatrix}$$

\$Subject

	(Intercept)	Days
308	2.2585509	9.1989758
309	-40.3987381	-8.6196806
310	-38.9604090	-5.4488565
330	23.6906196	-4.8143503
331	22.2603126	-3.0699116
332	9.0395679	-0.2721770
333	16.8405086	-0.2236361
334	-7.2326151	1.0745816
335	-0.3336684	-10.7521652
337	34.8904868	8.6282652
349	-25.2102286	1.1734322
350	-13.0700342	6.6142178
351	4.5778642	-3.0152621
352	20.8636782	3.5360011
369	3.2754656	0.8722149
370	-25.6129993	4.8224850
371	0.8070461	-0.9881562
372	12.3145921	1.2840221

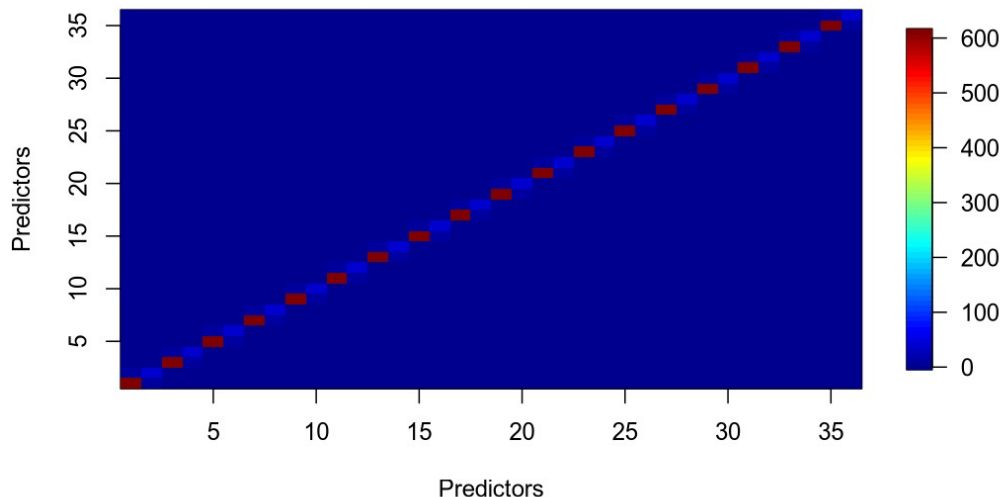
Q: What are the means of these columns?

with conditional variances for "Subject"

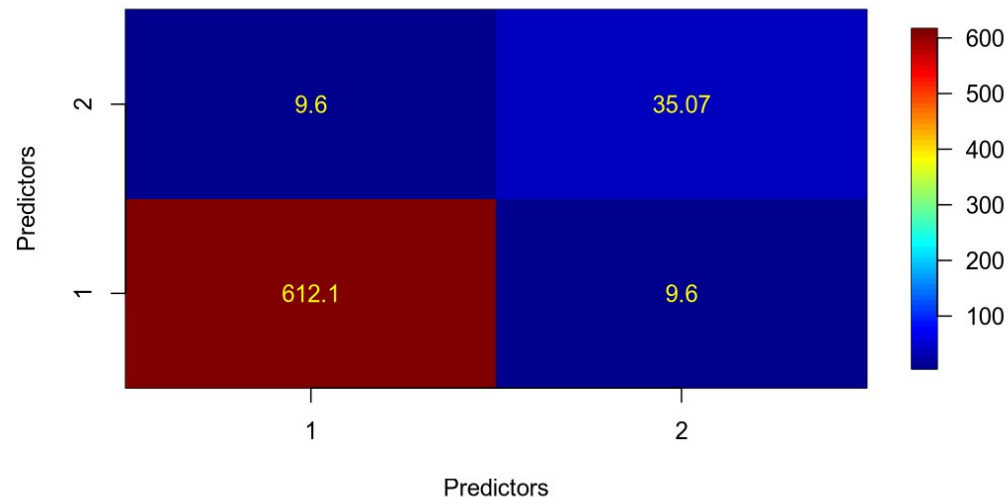
$$Corr = \frac{\hat{\Sigma}_{(1,2)}}{\sqrt{(\hat{\Sigma}_{(1,1)} \hat{\Sigma}_{(2,2)})}} = 0.07$$

$$\hat{\Sigma}_{\theta}$$

Variance-Covariance matrix



Variance-Covariance matrix (zoom)



Q: What would the plot look like if there was no covariance between predictors?

$$Corr = \frac{9.6}{\sqrt{(612.1 * 35.07)}} = 0.07$$

Why do we care about covariance?

Too much covariance between predictors,
and your model doesn't converge ... (G
cannot be inverted)

$$\begin{pmatrix} X^T X & X^T Z \\ Z^T Z & Z^T Z G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} X^T y \\ Z^T y \end{pmatrix}$$

Why do we care about covariance?

Too much covariance between predictors,
and your model doesn't converge ... (G
cannot be inverted)

... and even if you get convergence, the
estimated coefficients may be sensitive to
small changes in input

Example: covariance in predictors

```
get.SIGMA <- function(model) sigma(model)^2 * (getME(model, 'Lambda') %*% getME(model, 'Lambdat'))

for(seed in c(1, 7))
{
  print(paste('Random seed is:', seed))
  set.seed(seed)
  sleepstudy$Days2 <- 2 * sleepstudy$Days + rnorm(length(sleepstudy$Days))
  model <- lmer(Reaction ~ Days + Days2 + (Days + Days2 | Subject), data=sleepstudy)
  SIGMA <- get.SIGMA(model)
  print(paste('N columns is:', dim(SIGMA)[2]))
  print(paste('Estimated rank is:', rankMatrix(SIGMA)[1]))
}
```

```
## [1] "Random seed is: 1"
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## [1] "N columns is: 54"
```

```
## [1] "Estimated rank is: 36"
```

```
## [1] "Random seed is: 7"
```

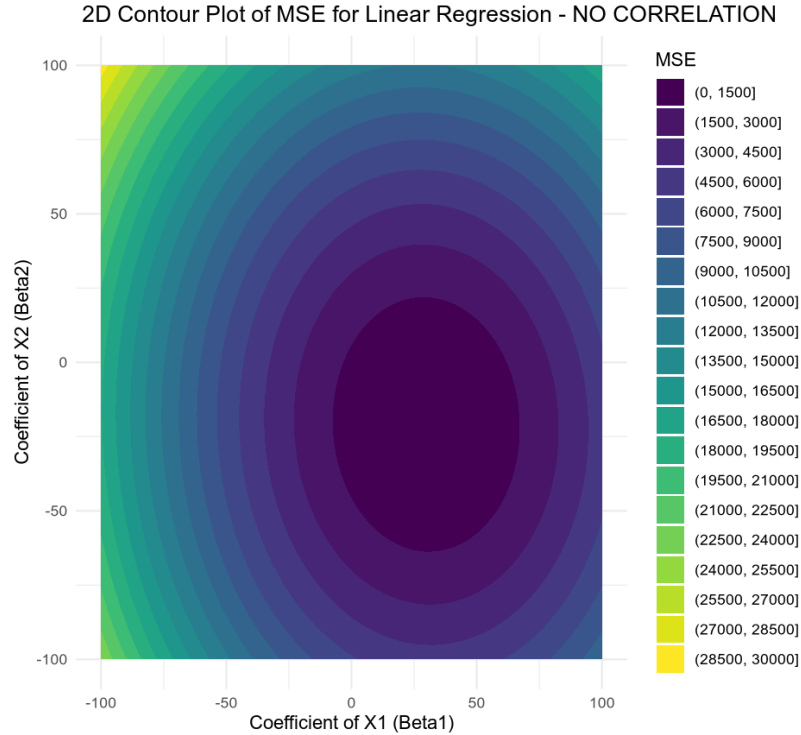
```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
```

```
## Model failed to converge with max|grad| = 0.0433554 (tol = 0.002, component 1)
```

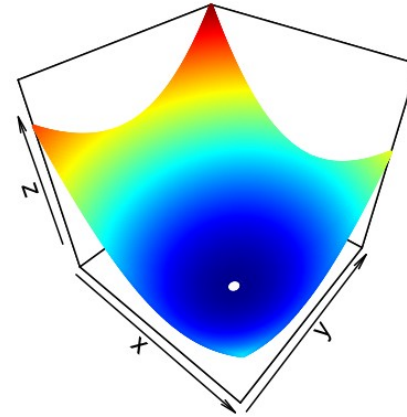
```
## [1] "N columns is: 54"
```

```
## [1] "Estimated rank is: 54"
```

True function: $y = 31x_1 - 21x_2$
 $x_1 = rnorm(100)$; $x_2 = rnorm(100)$

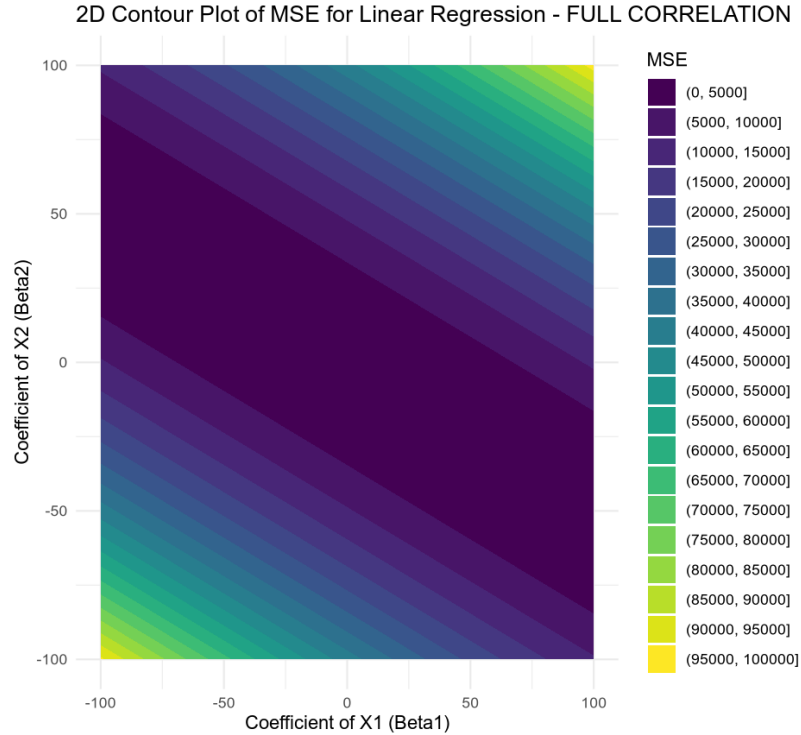


3D Contour Plot of MSE for Linear Regression - NO CORRELATION



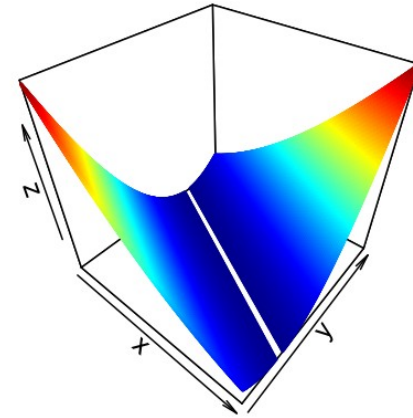
Full rank

True function: $y = 31x_1 - 21x_2$
 $x_1 = rnorm(100)$; $x_2 = 2x_1$



Singular

3D Contour Plot of MSE for Linear Regression - FULL CORRELATION



Hint for the future

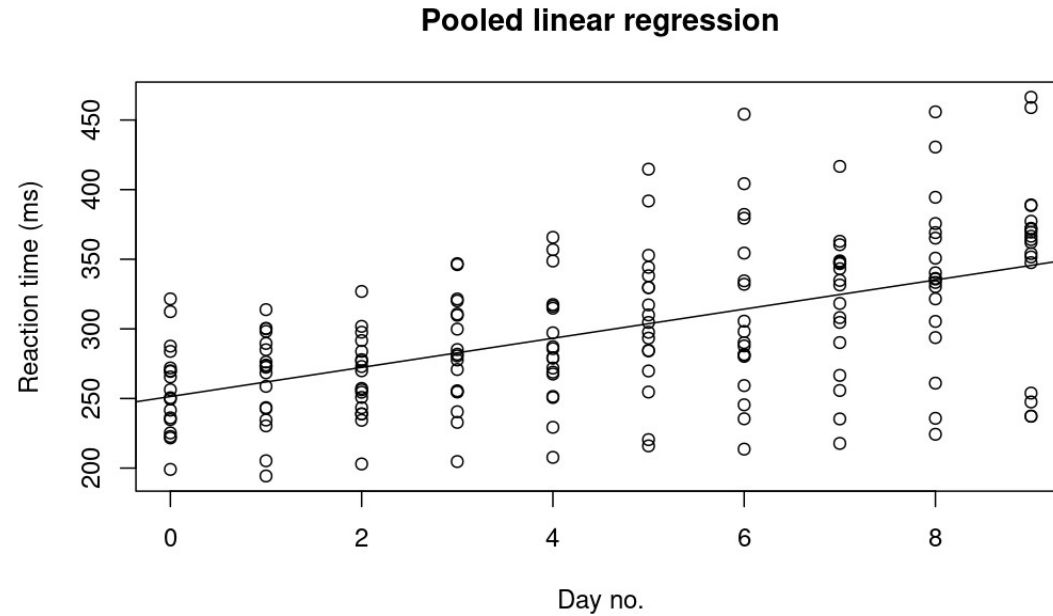
With “big data”, the variance-covariance matrices will matter a lot (machine learning)

Pooling

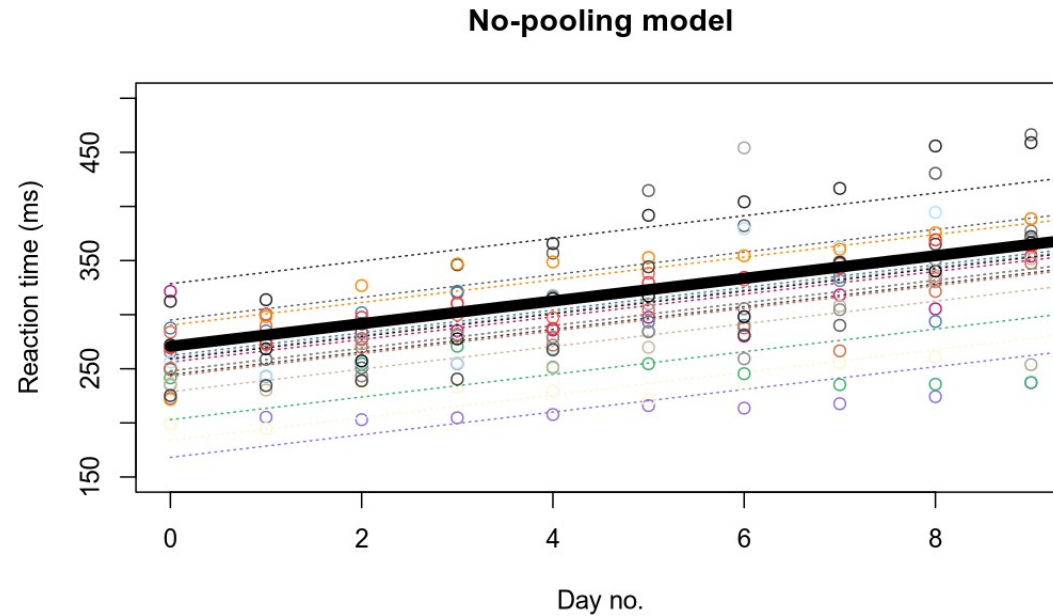
Overview – pooling

- Complete pooling
 - Ignores the categorical predictor, e.g. *Subject*, *altogether*
 - *lm(Reaction ~ Days)*
- No pooling
 - Overfits the categorical predictor, e.g. *Subject*, i.e. overstates the variation among *Subjects*
 - *lm(Reaction ~ Days * Subject - 1)*; models slopes and intercepts for each subject
 - *lm(Reaction ~ Days + Subject - 1)*; models intercepts for each subject
- Partial pooling
 - A compromise between the two extremes above. If a group, e.g. *Subject*, has few observations (high variance), it will be shrunk towards the overall mean. If *Subject* has many observations (low variance), it will be shrunk less towards the overall mean
 - *lmer(Reaction ~ Days + (Days | Subject) # models slopes and intercepts for each subject*
 - *lmer(Reaction ~ Days + (1 | Subject) # models intercepts for each subject*

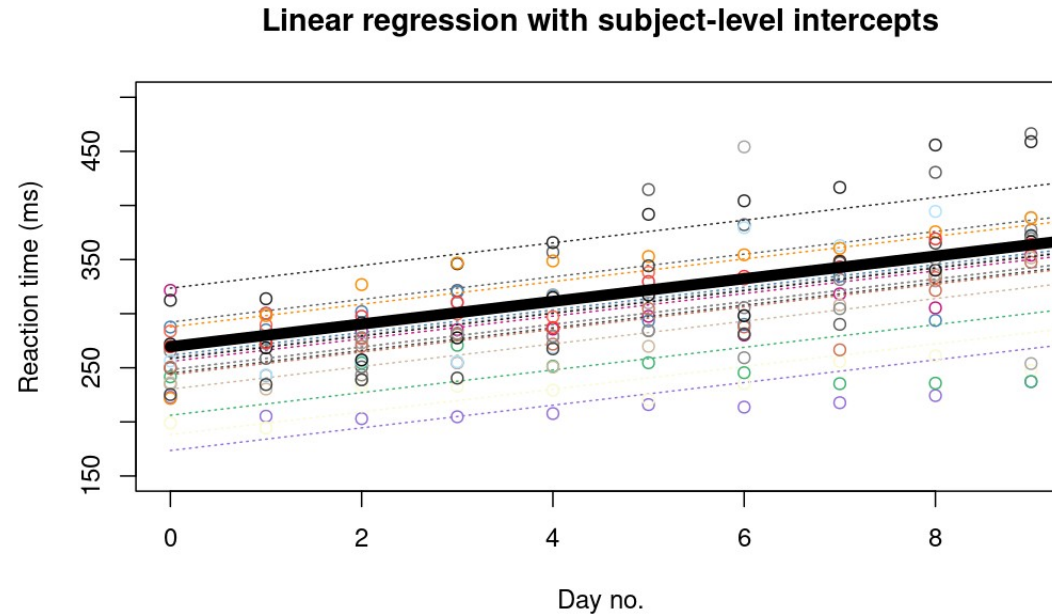
Complete pooling



No pooling



Partial pooling



Why is the shrinking small here?

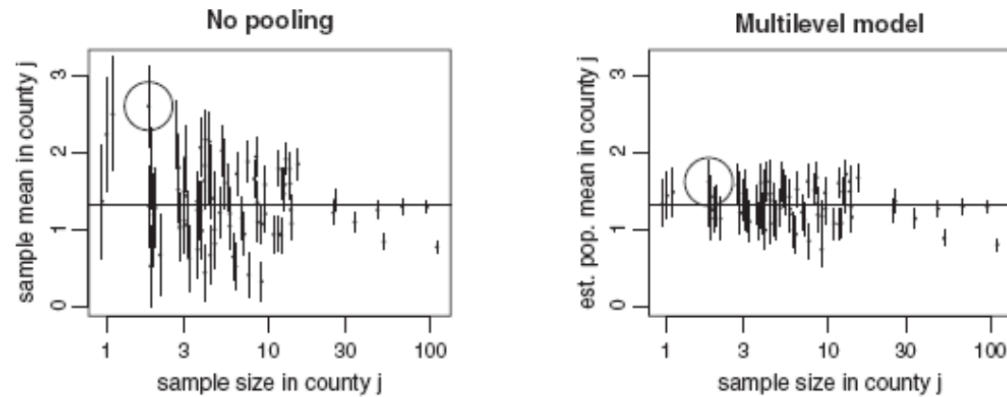


Figure 12.1 *Estimates $\pm$ standard errors for the average log radon levels in Minnesota counties plotted versus the (jittered) number of observations in the county: (a) no-pooling analysis, (b) multilevel (partial pooling) analysis, in both cases with no house-level or county-level predictors. The counties with fewer measurements have more variable estimates and larger higher standard errors. The horizontal line in each plot represents an estimate of the average radon level across all counties. The left plot illustrates a problem with the no-pooling analysis: it systematically causes us to think that certain counties are more extreme, just because they have smaller sample sizes.*

How is shrinking done?

HOW IS BIAS ADDED?

Penalised least squares:

$$r^2 = (y - X\hat{\beta} - Z\hat{u})^2 + \hat{u}^T G^{-1} \hat{u}$$

$(y - X\hat{\beta} - Z\hat{u})^2$ is the residual sum of squares

$\hat{u}^T G^{-1} \hat{u}$ is a penalty for 'spending' too much on u

Another way to look at shrinking

OR HOW TO ADD BIAS

$$\alpha_j \approx \frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} (\bar{y}_j - \beta \bar{x}_j) + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} \mu_\alpha \quad (12.4)$$

α_j : partially pooled response

$(\bar{y}_j - \beta \bar{x}_j)$: subject estimate of mean

μ_α : group estimate of mean

σ_y^2 : within-group variance

σ_α^2 : between-group variance

$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} + \frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} = 1$$

$$\frac{\frac{n_j}{\sigma_y^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} : \text{proportion assigned to subject estimate}$$

$$\frac{\frac{1}{\sigma_\alpha^2}}{\frac{n_j}{\sigma_y^2} + \frac{1}{\sigma_\alpha^2}} : \text{proportion assigned to group estimate}$$

Q's:

What happens to the estimated α_j ,
when respectively n_j , σ_y , or σ_α :

- 1) increases?
- 2) decreases?
- 3) is 0?
- 4) goes towards infinity?

Linear mixed model fit by REML ['lmerMod']
 Formula: Reaction ~ Days + (1 | Subject)

Data: sleepstudy
 REML criterion at convergence: 1786.465
 Random effects:

Groups	Name	Std.Dev.
Subject	(Intercept)	37.12
Residual		30.99

Number of obs: 180, groups: Subject, 18

Fixed Effects:

(Intercept)	Days
251.41	10.47

Call:
 lm(formula = Reaction ~ Days,
 data = sleepstudy)
 Coefficients:

(Intercept)	Days
251.41	10.47

$\hat{\sigma}_{\alpha}$
 $\hat{\sigma}_y$

$j = 308$

$n_{308} = \text{length}(y_{308}) = 10$

$\bar{y}_{308} = \text{mean}(y_{308})$

$\bar{x}_{308} = \text{mean}(x_{308})$

Call:
 lm(formula = Reaction ~ Days + Subject - 1,
 data = sleepstudy)
 Coefficients:

Days	Subject308	Subject309	...
10.47	295.03	168.13	...

$\hat{\mu}_{\alpha}$

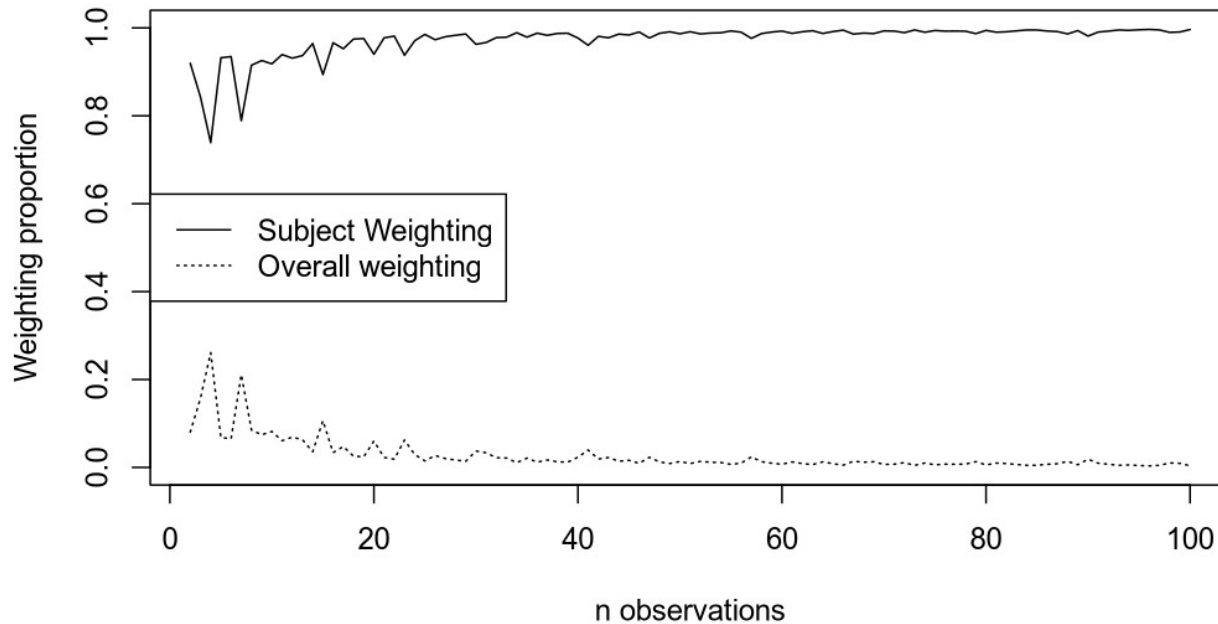
$\hat{\beta}$

$\bar{y}_{308} - \hat{\beta} \bar{x}_{308}$

When $\sigma_{\alpha} > \sigma_y$

100 SIMULATIONS

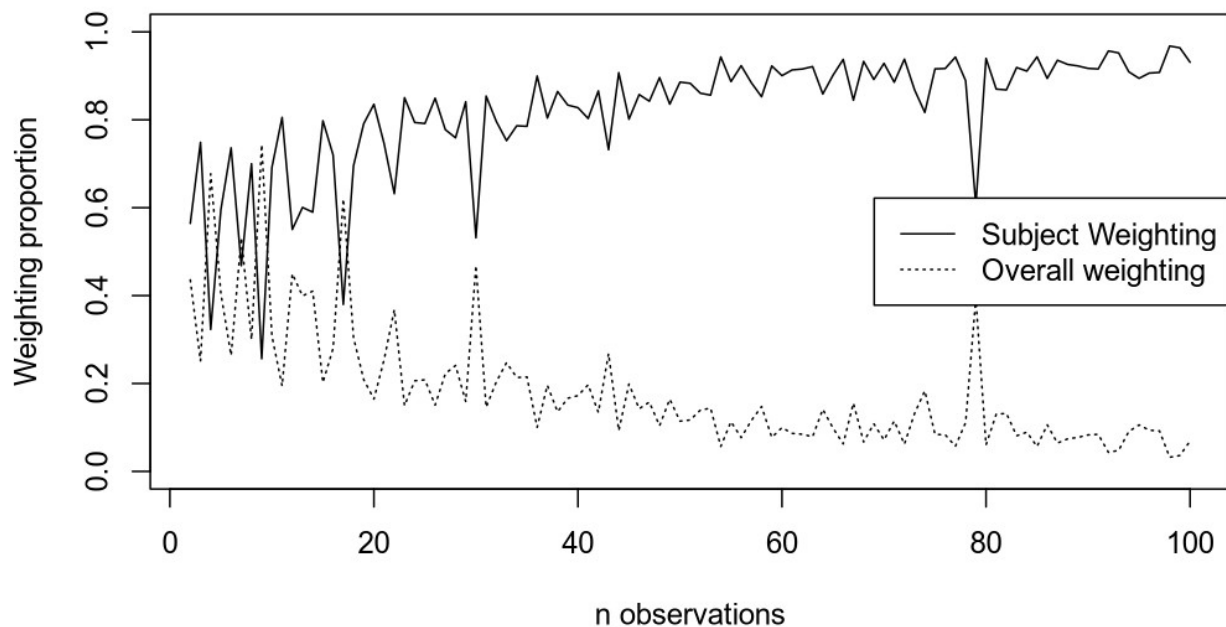
**Proportions as dependent on n observations
Between-sigma: 4, Within-sigma: 3**



When $\sigma_{\alpha} < \sigma_y$

100 SIMULATIONS

Proportions as dependent on n observations
Between-sigma: 4, Within-sigma: 10



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Student presentations:

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Next time –

Link functions and fitting generalised linear multilevel models

- Generalising general models
- Link functions
 - transforming from one scale to another
 - inverse functions
- Maximum likelihood estimation

Reading questions

- Chapter 5, Gelman & Hill
 - Why do we want to transform $X_i\beta$ using *logit*¹?
 - What is an odds ratio?
 - How are logistic regression coefficients interpreted?
- Sections 6.1 and 6.2 Gelman & Hill
 - What is the parameter, θ , of the Poisson distribution?
 - Why may overdispersion be an issue?